Machine Learning Data Preparation Report

Dataset: Telco Customer Churn

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Week 2: Data Preparation

1. GitHub Repository Link

Item	Details
Repository	ML-internship-Zee-outsourcing-task-2
Link	https://github.com/ayesha-tariq05/ML-internship-Zee-outsourcing-task-2

2. Dataset Overview

Attribute	Description
Source	https://www.kaggle.com/datasets/blastchar/telco-customer-churn
Total Rows	7,043
Total Columns	21 (initially)
Target Column	Churn (whether the customer left within the last month)
Problem Type	Binary Classification

3. Exploratory Data Analysis (EDA) and Insights

During the exploration phase, multiple charts were generated to understand the dataset characteristics:

- **Target Distribution Countplot:** Showed an imbalance in the dataset, with a higher proportion of non-churned customers than churned customers.
- **Tenure Histogram:** Revealed that a high concentration of customers either leave very early (low tenure) or stay long-term.
- **Correlation Heatmap:** Highlighted relationships between numerical variables such as tenure, MonthlyCharges and TotalCharges.
- **Contract vs. Churn Plot:** Indicated that customers on month-to-month contracts have a significantly higher churn rate than those on one-year or two-year contracts.

4. Data Cleaning Decisions

Column / Element	Problem Identified	Action Taken	Reason / Justification
customerID	Unique alphanumeric identifier	Dropped	Acts as an arbitrary row identifier and adds no predictive pattern for machine learning.
TotalCharges	Hidden missing values stored as blank white spaces (' ')	Converted to numeric (NaN), then median-imputed	Blank spaces blocked numerical operations; median imputation prevents outlier skewness.
Dataset Rows	Duplicate records present	Removed duplicates	Prevents model bias and data leakage during the train-test split.

5. Feature Engineering

Two new features were engineered to provide stronger predictive signals to the model:

- **Avg_Monthly_Charge:** Calculated by dividing TotalCharges by (tenure + 1). This captures historical spending relative to how long the customer has stayed with the company.
- **Is_Long_Term:** A binary indicator (1 if tenure > 24 months, else 0). This helps the model instantly distinguish long-term loyal customers from short-term ones.

6. Train/Test Split Strategy

Parameter	Setting
Split Ratio	80% training set, 20% testing set (random_state=42)
Stratification	stratify=y: both sets keep the same churn / non-churn ratio, preventing class imbalance distortion
Class Proportions	~73.5% non-churn and ~26.5% churn maintained across both subsets

7. Preprocessing and Data Leakage Prevention

To prepare the data for machine learning algorithms without leaking test information into training:

- **Numerical Features:** Missing values filled with the median, followed by scaling with StandardScaler to bring all features to a standard normal distribution.
- **Categorical Features:** Missing values filled with the most frequent value, followed by encoding with OneHotEncoder.
- **ColumnTransformer & Pipeline:** Both transformations were wrapped into a single ColumnTransformer.

- **Data Leakage Avoidance:** The preprocessor was fitted exclusively on X_train (fit_transform) and only applied to X_test (transform). No test set statistics were used during training preparation.

8. Final Results (Dataset Shapes)

After preprocessing, the final dimensions of the datasets were confirmed as:

Dataset	Shape
X_train_processed	(5634, [transformed feature count])
X_test_processed	(1409, [transformed feature count])
y_train	(5634,)
y_test	(1409,)

Missing Values Check: Confirmed zero missing values (NaN) remaining in both X_train_processed and X_test_processed.

9. Challenges Faced and What Was Learned

Challenge: Dealing with hidden missing values in TotalCharges, where empty strings (' ') caused Python to treat the numerical column as an object (text) type.

Learning: Gained hands-on experience in robust error handling using pd.to_numeric(..., errors='coerce'), building automated preprocessing pipelines with scikit-learn's ColumnTransformer, and strictly enforcing data isolation to eliminate data leakage.